



Module Overview

This module introduces the fundamental concepts and algorithms of machine learning, covering supervised, unsupervised, and reinforcement learning paradigms. Students will learn about training and testing data, overfitting and underfitting, biases in AI, and model evaluation metrics. The module covers some classical algorithms, including decision trees, linear regression, and clustering algorithms, providing hands-on experience with real datasets.



Learning Objectives

By the end of this module, you should be able to:

- Differentiate between supervised, unsupervised, and reinforcement learning
- Evaluate classical machine learning algorithms
- Apply appropriate evaluation metrics for classification and regression tasks
- Diagnose and address overfitting using regularization and cross-validation
- Select appropriate algorithms for different types of problems



Task List

- Participate in the weekly discussion board
- Solve and submit the weekly assignment
- Additional reading:
 - Aggarwal, C. C., & Reddy, C. K. (2013). Data clustering: algorithms and applications. CRC Press.
 - Barocas, S., Hardt, M., & Narayanan, A. (2019). Fairness and machine learning. fairmlbook.org.
 - Bengio, Y., LeCun, Y., & Hinton, G. (2021). Deep learning for AI

